

Ahmed Mohamed

AI Engineer · Technical Lead

I build LLM systems that go into production and stay there: extraction pipelines, multi-provider structured output, cell-level accuracy evaluation, and the AWS and data-governance setup to run them on sensitive data. Data scientist by training (PhD, 30+ papers, statistics and ML on a 30,000-sample study) and a backend engineer by practice (Go services and financial reporting for an institutional Ethereum operator).

EXPERIENCE

Co-founder & Technical Lead

AIWhisperers, Australia

2023 - now

- ▶ Built production LLM systems for SMEs, including a human-reviewed platform converting supplier contracts into structured commercial data for a B2B travel company.
- ▶ Wrote the company's LLM tooling: multi-provider structured-output library, composable extraction chains, LLM-run caching, cell-level evaluation comparing prompt strategies and models; DSPy prompt optimisation.
- ▶ Established the PHI data-governance framework (Privacy Act 1988, ISO 27001-aligned) and secure AWS architecture: private-subnet SageMaker inference, KMS-encrypted RDS/S3, least-privilege IAM.
- ▶ Delivered LLM-for-business workshops for companies in retail, health, tourism and mining.

LLM systems

Evals

Data governance

AWS

Senior Software Engineer

Bitwise Onchain Solutions (ex-Attestant)

2022 - now

- ▶ Backend engineer for institutional Ethereum and multi-chain staking services.
- ▶ Own the client reporting and invoicing system with multi-currency support; maintain client-facing APIs and the multi-chain staking dApp.

Go

Ethereum

Financial systems

Bioinformatician, Senior Research Officer

WEHI, Melbourne

2020 - 2025

- ▶ Led experimental design, QC, statistics and ML for the ASPREE proteomics programme: 12,138 participants, ~30,000 LC-MS runs, >10 TB – at the time 10× the largest published MS population proteomics study.
- ▶ Built the Nextflow processing pipeline on SLURM HPC (DIA-NN, out-of-memory HDF5 matrices) and a BigQuery-backed QC dashboard watching hundreds of runs a day.
- ▶ Designed the analysis-first experimental layout that made batch correction and longitudinal modelling work at scale; ran LASSO biomarker models across 13 conditions.
- ▶ Built vissE.cloud end-to-end (React/TypeScript, Flask, Redis, Docker): a public gene-set analysis web server across bulk, single-cell and spatial data; first author of the NAR paper.
- ▶ Senior author on an ML urine-spectroscopy test for bladder-cancer recurrence (ROC > 0.8).

Proteomics

Statistics & ML

Nextflow

BigQuery

Research Officer

QIMR Berghofer & UQ Diamantina, Brisbane

2017 - 2020

- ▶ Created lipidr, an R/Bioconductor package for lipidomics analysis (160+ citations), and LipidSuite, its web server for non-programmers; sole developer of RaftProt V2 – tools accessed tens of thousands of times.
- ▶ Statistical lead for collaborator studies; taught R package development and Software Carpentry Python; associate supervisor of a PhD student to completion.

Lipidomics

Bioconductor

Biostatistics

PhD · JSPS Visiting Fellow

Kyoto University · Boston University

2012 - 2016

- ▶ NetPathMiner, an R/Bioconductor package for network path mining (Bioinformatics); NMRPro, interactive web processing of NMR spectra (Bioinformatics).
- ▶ Pan-cancer analysis of somatic mutations in protein complexes.

Network biology

Cancer genomics

✉ ahmed@tailor.example

🐙 github.com/ahmohamed

🌐 linkedin.com/in/amfy10

📍 Melbourne, Australia

EDUCATION

PhD, Bioinformatics & Chemical Genomics

Kyoto University, Japan · 2012-2016

MSc, Advanced Pharmaco-science

Osaka University, Japan · 2010-2012

BPharm, Pharmaceutical Sciences

Suez Canal University, Egypt · 2002-2007

SKILLS

LLM engineering

Extraction pipelines, structured output · Cell-level evals, DSPy prompt optimisation · RAG, multi-provider tooling

Data science

Statistics & ML at population scale · Cancer genomics, Bioconductor · lipidr, NetPathMiner, vissE (author)

Software & infra

Python, Go, TypeScript/React · AWS, Kubernetes, Docker, PostgreSQL · Nextflow, SLURM HPC

Web3

Ethereum staking services · Financial reporting & invoicing

AWARDS

🏆 ICR Award for Graduate Students, 2015

🏆 Winner, Novartis BioCamp, 2013

🏆 JSPS International Training Program Fellowship, 2013

🏆 Rotary Yoneyama Memorial Scholarship

PUBLICATIONS

30+ papers. NetPathMiner (Bioinformatics), NMRPro (Bioinformatics), lipidr (160+ citations, in Bioconductor since 2019), vissE.cloud (NAR, first author).

LANGUAGES

English, Arabic, Japanese

PROJECTS

Tailor2026

This CV – career knowledge base + agentic tailoring, open source

- Harvest your career once; tailor CVs to any job in minutes. This document is its output.

Claude Code typst

Notebrainer2025

Provenance-first agent memory · Claude Code plugin

- Memory layer where every stored fact traces to its source text.

Postgres pgvector

MapMyMath2024

Procedural maths game · web, iOS, Android

- 1,320 catalogued exam questions; built solo in five weeks.

Python React/WebGL